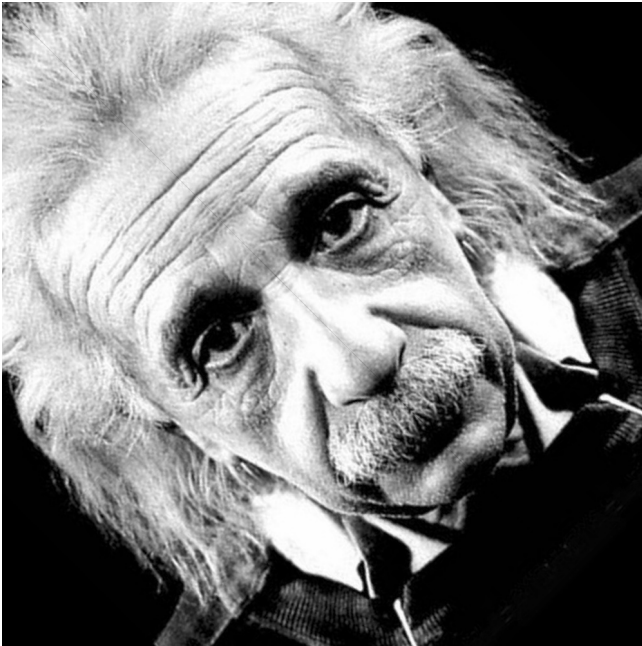




- The size of this image is 1000×1000 pixels.
- The color of each pixel is represented by an integer between 0 (black) and 255 (white).
- The whole image is described by a (symmetric) matrix A consisting of $1000 \times 1000 = 1,000,000$ numbers
- Each number is stored in 1 byte, so the image file size is 1,000,000 bytes (≈ 1 MB).

How to make the image file smaller:

1) Find the spectral decomposition of the matrix A :

$$A = \lambda_1(\mathbf{u}_1\mathbf{u}_1^T) + \lambda_2(\mathbf{u}_2\mathbf{u}_2^T) + \dots + \lambda_{1000}(\mathbf{u}_{1000}\mathbf{u}_{1000}^T)$$

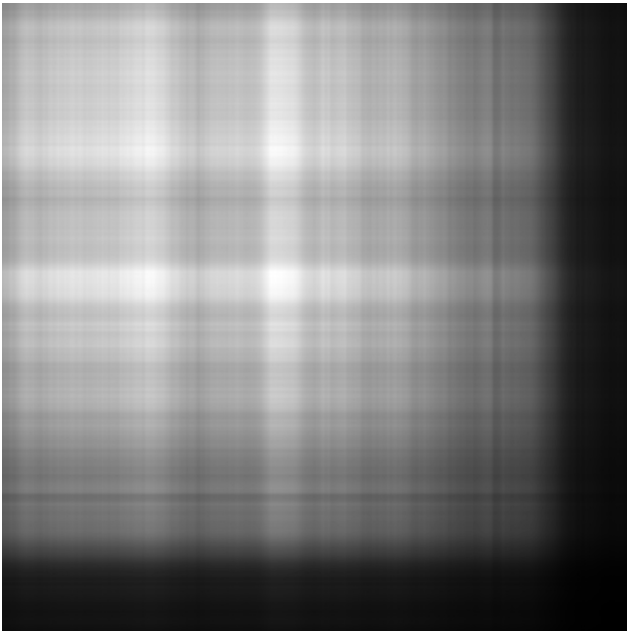
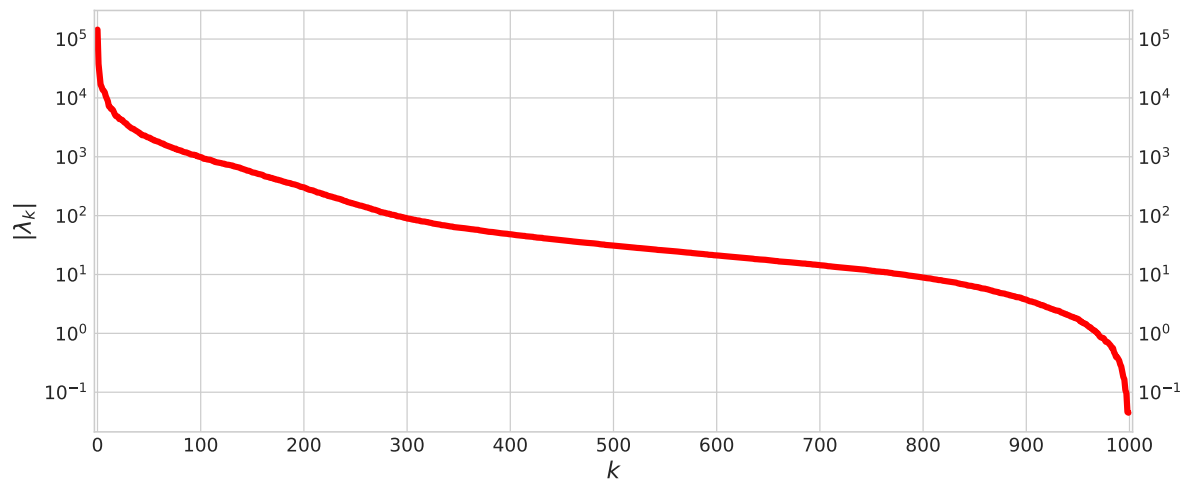
where $|\lambda_1| \geq |\lambda_2| \geq \dots \geq |\lambda_{1000}|$.

2) For $k = 1, \dots, 1000$ define:

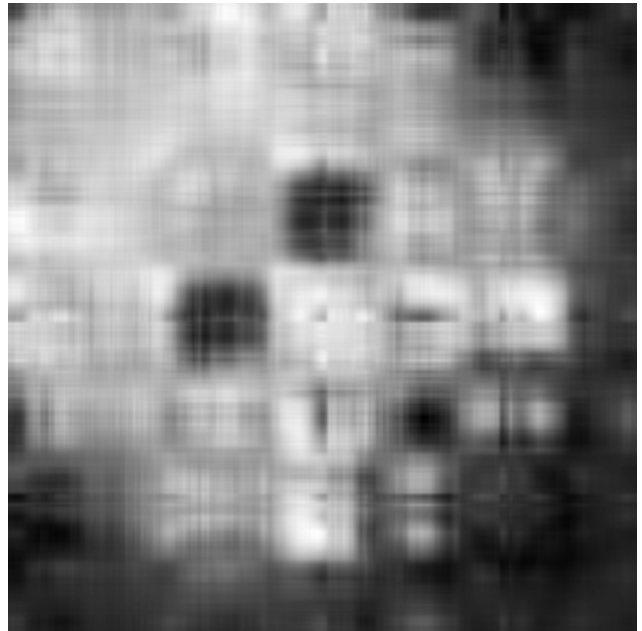
$$B_k = \lambda_1(\mathbf{u}_1\mathbf{u}_1^T) + \lambda_2(\mathbf{u}_2\mathbf{u}_2^T) + \dots + \lambda_k(\mathbf{u}_k\mathbf{u}_k^T)$$

This matrix approximates the matrix A and can be stored using $k \cdot (1000 + 1)$ numbers (i.e. $k \cdot (1000 + 1)$ bytes).

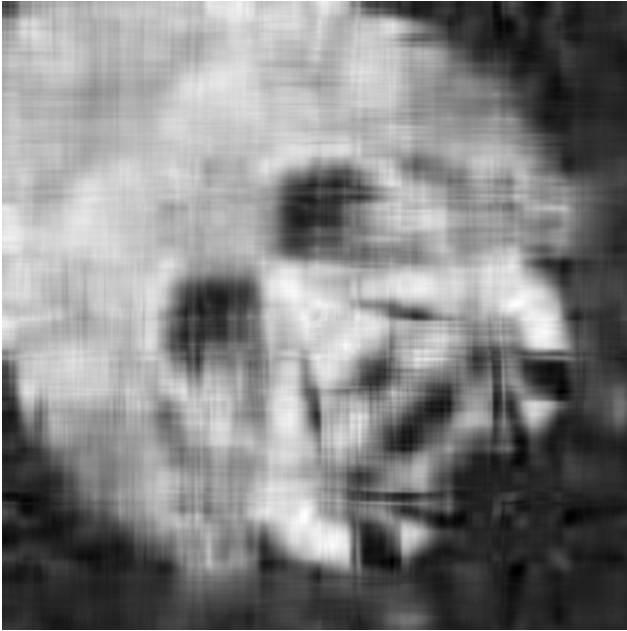
Eigenvalues of the matrix A



matrix B_1
1001 bytes
compression 1000:1



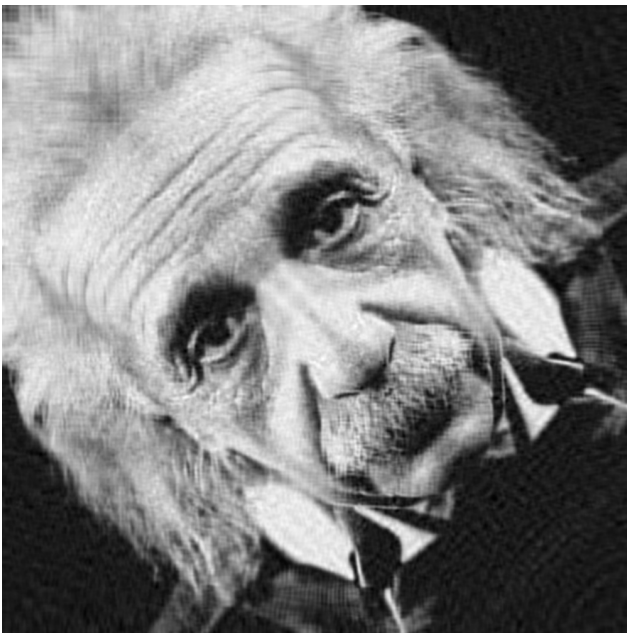
matrix B_5
5005 bytes
compression 200:1



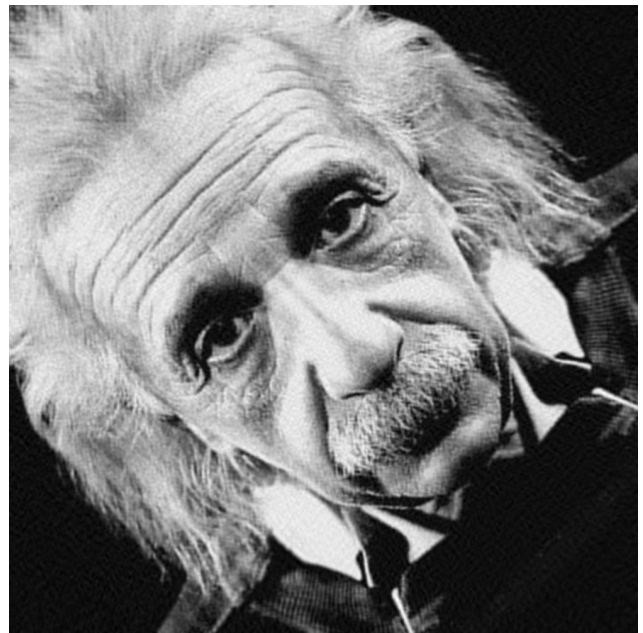
matrix B_{10}
 10,010 bytes
 compression 100:1



matrix B_{20}
 20,020 bytes
 compression 50:1



matrix B_{50}
 50,050 bytes
 compression 20:1



matrix B_{100}
 100,100 bytes
 compression 10:1